CP-450 Series Horizontal Centrifugal Pump

Model Series: CP-450 | Equipment Type: Horizontal Centrifugal Slurry Pump

1. Overview

The CP-450 Series is a horizontal, end-suction centrifugal pump designed for heavy duty slurry transfer in mineral processing circuits. It features a replaceable elastomer or hard-metal liner, a mechanical seal or gland packing stuffing box (model-dependent), and a heavy-duty bearing frame for continuous duty.

2. Technical Specifications

Parameter	Value
Flow range	50 – 1,800 m³/h
Max head	65 m
Casing material	28% chrome white iron (standard)
Impeller	28% chrome white iron, semi-open, 2- or 3-vane
Shaft seal	Gland packing (standard) or mechanical seal (CP-450-MS)
Bearing frame lubrication	Grease (standard), oil-mist (high-duty option)
Max particle size	Up to 25mm depending on impeller clearance setting
Drive	V-belt or direct-coupled via flexible coupling
Max continuous speed	1,800 rpm

3. Installation Guidelines

1. Mount pump base on a rigid, level foundation; check baseplate with a precision level before grouting.
2. Align pump and motor shafts within 0.05mm TIR using a laser or dial-indicator method before final coupling bolt tightening.
3. Connect suction piping with a minimum straight run of 5 pipe diameters before the suction flange to avoid pre-swirl.
4. Set impeller-to-liner clearance per the clearance gauge supplied with the pump — see Section 4 for the adjusting-nut torque spec.
5. Prime the pump fully before start-up; never dry-run a centrifugal slurry pump even briefly.
6. Verify rotation direction against the arrow cast on the casing before coupling to the driver.

4. Torque Specifications

Apply torque values below using a calibrated torque wrench. Over-torquing bolted connections can distort sealing faces and cause premature gasket or packing failure.

Fastener / Location	Size	Torque (N·m)
Casing bolts, CP-450-50 to -200	M16	120
Casing bolts, CP-450-250 to -450	M20	200
Impeller retaining nut	M36	450
Bearing housing cap screws	M12	60
Coupling guard fasteners	M8	18
Impeller clearance adjusting nut (locknut)	M24	90

5. Maintenance Schedule

Interval	Task
Daily	Check bearing temperature (alarm at 80°C) and listen for abnormal cavitation noise.
Weekly	Grease bearing frame per lubrication chart; inspect gland/seal for excessive leakage.
Monthly	Re-check and adjust impeller-to-liner clearance as wear progresses; record clearance in the maintenar
Quarterly	Inspect coupling alignment and elastomer insert condition.
Annually or at overhaul	Full strip-down; inspect/replace liner, impeller, shaft sleeve, and bearings regardless of apparent condit

6. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Loss of discharge pressure / reduced flow	Impeller wear or excessive impeller-to-liner clearance	Reset clearance per Section 4; if already at limit, replac
Excessive bearing housing vibration	Impeller imbalance from wear, or misalignment	Check coupling alignment first; if aligned, inspect impeller fo
Pump will not prime / loses prime repeatedly	Air ingested at suction side, or worn mechanical seal/gland	Check suction piping joints and foot valve; inspect seal/pac
Overheating at bearing frame	Over- or under-greasing, or bearing nearing end of life	Verify grease quantity against lubrication chart; replace bea
Pump seal / packing weeping excessively	Packing worn or incorrectly tensioned	Adjust gland follower slightly; if adjustment no longer helps, repack or replace mechanical seal set if a

7. Safety Warnings

■ Never operate the pump with the coupling guard removed.

■ Never dry-run the pump, even momentarily — this is the leading cause of mechanical seal and bearing failure in slurry service.

■ Lock out and tag out the driver before any clearance adjustment, liner, or impeller work.

■ Hot surfaces: bearing housings can exceed 70°C in normal operation — allow cooling before hands-on inspection.